

Haider Riaz Khan

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EDUCATION

2024 – 2026 **M.S. in Mathematics** CITY COLLEGE OF NEW YORK
Research Project: Coding of Piecewise Smooth Dynamical Systems (Advised by Tamara Kucherenko)

2017 – 2018 **M.A. in Philosophy** UNIVERSITY OF WATERLOO
 Thesis: The Phenomenological Origins of Property (Advised by John Turri)
 Seminars: Self-Knowledge, Human & Computer Intelligence, and Foundations of Quantum Theory

2010 – 2015 **B.Sc. in Physics and Computer Science** MCGILL UNIVERSITY
Special Research Project: The Building of a Cellular Detector (Advised by Paul François)

ENGINEERING EXPERIENCE

2025 - Pres **Co-Director and Lead Engineer** RED CALENDAR (NYC, USA)

In collaboration with [Theo Ellin Ballew](#), I lead and maintain the development of a curated web calendar for arts and action events in NYC. Events can be added by anyone, subject to editor approval.

2024 - Pres **Software Engineer** BASE EDUCATION (NYC, USA)

Wrote GraphQL APIs and built out the application logic for delivering data insights—including computing student-level data, aggregate statistics, and advanced features such as differentially private statistics. Created fast, efficient, and secure data pipelines that power the backend systems.

2021 - 2022	Senior Software Engineer	AVENUE 8 (NYC, USA)
	Implemented Avenue 8's observability infrastructure. Wrote automation, monitoring, and alerting scripts. Automated the management of AWS resources using Terraform. Improved network security.	

2019 - 2021 **Software Engineer** LIFION BY ADP (NYC, USA)

Built Lifion's graph database framework using a microservices architecture, Node.js, Gremlin, and distributed message queuing services. Wrote optimized Gremlin queries and SQL subroutines.

2016 - 2017	Software Engineer	MEDIDATA SOLUTIONS (NYC, USA)
Contributed to the development of a Java-based combinatorial test data generator called BIT-TAG.		

AWARDS & HONORS

2025 Dr. Barnett and Jean Hollander Rich Graduate Scholarship CITY COLLEGE

2025 Dr. Barnett and Jean Hollander Rich Summer Fellowship CITY COLLEGE

2018 Paul Seligman Memorial Scholarship UNIVERSITY OF WATERLOO

2015 Dean's Multidisciplinary Undergraduate Research List MCGILL

2014 NSERC-CREATE Neuroengineering Award MCGILL

Alexander Rutherford Scholarship

PUBLICATIONS

CONFERENCE AND JOURNAL ARTICLES

1. **Khan, Haider Riaz**, & John Turri. “Phenomenological Origins of Psychological Ownership.” *Review of General Psychology* 26, no. 4 (December 2022): 446–63.
2. Li, Nan, Yu Lei, **Haider R. Khan**, Jingshu Liu, and Yun Guo. “Applying Combinatorial Test Data Generation to Big Data Applications.” In *Proceedings of the 31st IEEE/ACM International Conference on Automated Software Engineering (ASE ’16)*, 637–47. New York, NY: Association for Computing Machinery, August 2016.

ESSAYS

1. “The Forever War.” *Commune*, Issue 5, Winter 2020.

RESEARCH & TEACHING

2018	Translator	PHILOSOPHICAL SCIENCE LAB, UNIVERSITY OF WATERLOO
	Translated English stimuli into Pashto and Urdu for the purposes of a cross-cultural study of the “ought implies can” principle. Provided feedback on research paper drafts to Prof. Turri.	
Winter 2018	Teaching Assistant	UNIVERSITY OF WATERLOO
	Biomedical Ethics (Prof. Andrew Stumpf)	
Fall 2017	Teaching Assistant	UNIVERSITY OF WATERLOO
	Professional and Business Ethics (Dr. Jim Jordan)	
2014 - 2015	NSERC Neuroengineering Fellow	RUTHAZER LAB, MONTREAL NEUROLOGICAL INSTITUTE
	Wrote CANDLE-J, an open-source 3D image denoising software designed as an ImageJ plugin. CANDLE-J is adept at processing deep in vivo 3D multiphoton microscopy images where the signal-to-noise ratio (SNR) is low. It is written in Python, Java, and multithreaded C.	
2012 - 2014	Computational Scientist	COOK LAB, MCGILL UNIVERSITY
	Developed MATLAB routines to measure the cross-correlation of microsaccades and microstimulations (in area MT of the visual cortex). Created a library of MATLAB functions to compute joint metrics of neural activity. The joint metrics are computed for both single-unit and multi-unit neuronal spikes.	
Fall 2011	SUS Peer Tutor	MCGILL UNIVERSITY
	Met with students routinely to go over physics and chemistry concepts; held exam review sessions.	

TALKS & PRESENTATIONS

2016	comBinatorial bIg daTa Test dAta Generator (BIT-TAG)	ASE'16, SINGAPORE
2014	Neuroscience From Below	STUDENT SUMMER COLLOQUIUM, MCGILL

TECHNICAL SKILLS

Programming Languages: JavaScript/TypeScript, Clojure, Java, Python, C, Bash, SQL
Data, Storage & Querying: MySQL, Amazon S3, GraphQL, Elasticsearch, Redis
Development & Cloud: Git, GitHub Actions, Docker, AWS Lambda
Scientific Computing & Build: MATLAB, Jupyter, SageMath, GCC, javac

LANGUAGES

English, Urdu, Pashto